ARIA: Agentic Research Intelligence Assistant

Technical Product Requirements Document

KIIT University · B.Tech CS Capstone April 2026

Domain: Agentic AI / LLM Systems
Capstone Type: LangGraph + RAG + Agentic Orchestration
Primary Stack: Python · LangGraph · ChromaDB · Streamlit
Deployment Target: Streamlit Cloud / HuggingFace Spaces
Difficulty: Medium-Hard (extends base curriculum)
Timeline: 4–5 weeks build + 1 week polish

01 Abstract

ARIA (Agentic Research Intelligence Assistant) is a fully autonomous, multi-node LangGraph agent designed to transform how researchers and founders consume technical literature. A user states a research question in plain language; ARIA autonomously plans a multi-step investigation, decomposes it into sub-queries, retrieves grounded context from a ChromaDB knowledge base, executes live web and arXiv searches in parallel, synthesises a structured report, evaluates its own faithfulness via a self-reflection gate, and iterates if quality falls below threshold — all within a stateful, multi-turn conversation session.

The project satisfies all six mandatory capstone capabilities prescribed by the course (LangGraph Workflow, RAG Knowledge Base, Conversation Memory, Self-Reflection, Tool Use, and Deployment) and extends them through a Plan-Execute orchestration pattern, a domain-specific arXiv tool, and a citation-aware Streamlit report UI.

02 Problem Statement & Motivation

The volume of AI/ML literature has grown at an exponential rate since 2020. Researchers, students, and startup founders face three compounding problems:

Information Overload arXiv alone publishes 500+ ML papers per day. Manual triage is impossible at this velocity. Founders and researchers spend 40–60% of their research time on literature discovery rather than synthesis.

Grounding Gap Existing LLM chatbots hallucinate citations and fabricate paper titles. Without a retrieval-grounded architecture, AI-assisted research cannot be trusted for academic or investment decision-making.

Context Fragmentation Each research query starts from scratch. There is no system that maintains session context, remembers prior findings, and allows iterative drill-down — the way a human research assistant would.

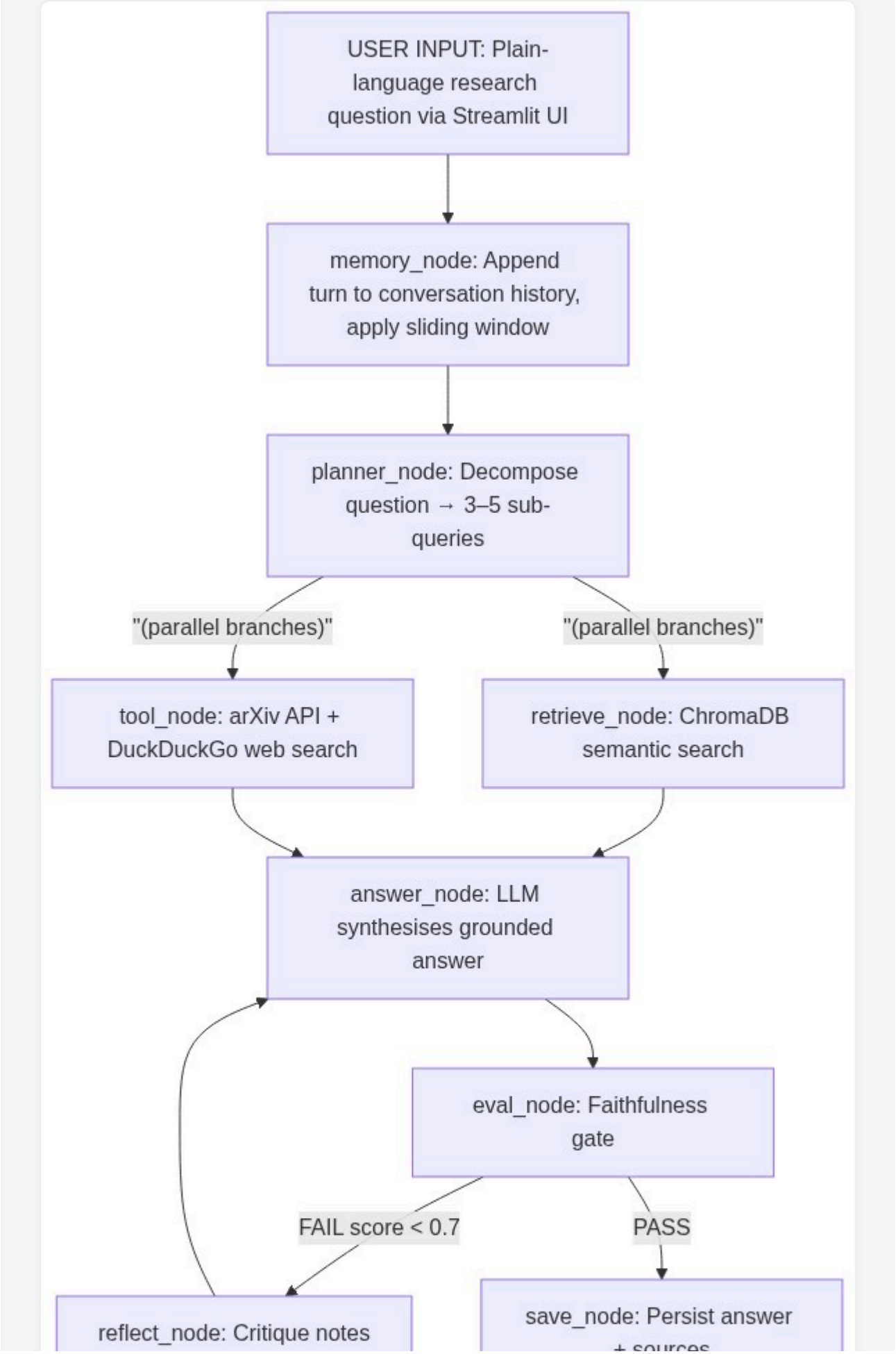
Why this matters: ARIA is not a hypothetical tool — it directly accelerates real work and demonstrates the capacity to build production-grade agentic systems.

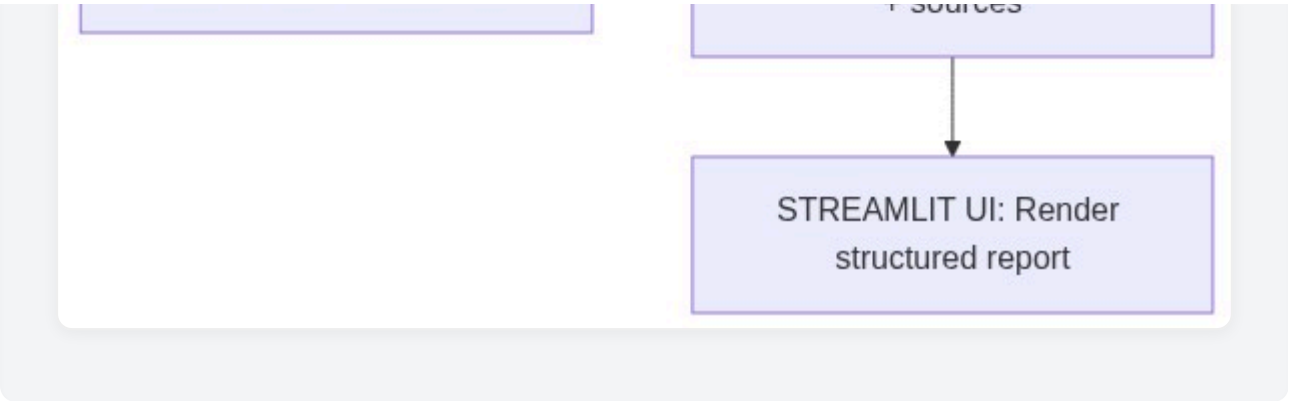
03 Project Goals & Success Criteria

#	Goal	Success Criterion	Priority
G1	Autonomous research planning	Agent decomposes any question into 3–5 sub-queries without user intervention	P0
G2	Grounded answers only	100% of answers cite at least one ChromaDB or arXiv source	P0
G3	Self-correcting quality gate	Faithfulness score gates all answers; retries triggered when score < 0.7	P0
G4	Multi-turn memory	Agent correctly references facts from 3+ prior turns in the same session	P0
G5	Live web + arXiv retrieval	arXiv tool returns real paper metadata; DuckDuckGo returns live snippets	P1
G6	Non-technical deployment	Any user can run a full research query via Streamlit without touching code	P0
G7	Structured report output	Every answer is formatted as: Summary, Key Findings, Sources, Follow-ups	P1
G8	Domain configurability	ChromaDB knowledge base is swappable by domain via config.yaml	P2

04 System Architecture Overview

ARIA implements a Plan-Execute-Reflect-Synthesise (PERS) agentic pattern on top of LangGraph's StateGraph API. Unlike simple RAG chains, the architecture has a planning phase that decomposes queries before any retrieval occurs, enabling parallel sub-query execution and genuinely multi-source synthesis.


USER INPUT: Plain-language research question via Streamlit UI
memory_node: Append turn to conversation history, apply sliding window
planner_node: Decompose question → 3–5 sub-queries
"(parallel branches)"
"(parallel branches)"
tool_node: arXiv API + DuckDuckGo web search
retrieve_node: ChromaDB semantic search
answer_node: LLM synthesises grounded answer
eval_node: Faithfulness gate
FAIL score < 0.7
PASS
reflect_node: Critique notes
save_node: Persist answer



Key Architectural Decision: Plan-Execute Pattern Unlike simple router → retrieve → answer chains, ARIA inserts a `planner_node` that uses structured LLM output to decompose queries before touching any retrieval. This enables parallel sub-query execution, dramatically improves coverage on complex questions, and is the single biggest differentiator vs. other student projects in this cohort.

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05 LangGraph State Design

The `TypedDict` State is the contract between all nodes. It must be designed before any node is written. Every field that a node reads or writes must appear here.

```
# aria/state.py
from typing import TypedDict, List, Optional, Annotated
import operator

class ARIAState(TypedDict):
    # Input
    question:        str                    # current user question
    thread_id:       str                    # conversation session ID

    # Planning
    sub_queries:     List[str]              # planner decomposition
    route:           str                    # 'retrieve' | 'tool' | 'both'

    # Retrieval
    retrieved:       str                    # ChromaDB context chunks
    sources:         List[str]              # source document names
    tool_result:     str                    # arXiv + web search results

    # Generation
    answer:          str                    # synthesised LLM response
    report:          dict                   # structured report sections

    # Evaluation
    faithfulness:    float                  # quality score 0.0–1.0
    eval_retries:    int                    # safety valve counter
    reflection_note: str                    # critique from reflect_node

    # Memory
    messages:        Annotated[List[dict], operator.add]  # full history
    context_window:  List[dict]             # sliding window (last k=10)
```

Field	Type	Owner Node(s)	Purpose
`question`	str	entry / memory_node	User's current input
`thread_id`	str	entry	Session identifier for MemorySaver
`sub_queries`	List[str]	planner_node	Decomposed sub-questions
`route`	str	planner_node	Routing decision for parallel branch
`retrieved`	str	retrieve_node	ChromaDB concatenated chunks
`sources`	List[str]	retrieve_node	Provenance list for citation
`tool_result`	str	tool_node	arXiv + web concatenated results
`answer`	str	answer_node / reflect_node	Current best answer
`report`	dict	answer_node	Structured sections dict
`faithfulness`	float	eval_node	Grounding quality 0–1
`eval_retries`	int	eval_node	Max=2 before force-pass
`reflection...`	str	reflect_node	Critique for re-generation
`messages`	List[dict]	memory_node / save_node	Full conversation log
`context_win...`	List[dict]	memory_node	Sliding window k=10 turns

06 Node Specifications

- **memory_node**
- Reads: question, messages, thread_id
- Writes: messages (append), context_window (sliding window)

- Logic: Append current question as HumanMessage. Trim messages to last k=10 pairs. Key rule: MUST run first on every turn.

- **planner_node**

- Reads: question, context_window
- Writes: sub_queries (List[str]), route (str)

- Logic: Prompt LLM with structured output parser to produce 3–5 sub-queries. Route decision: 'retrieve', 'tool', 'both'.

- **retrieve_node**

- Reads: sub_queries, route
- Writes: retrieved (str), sources (List[str])

- Logic: For each sub-query, query ChromaDB collection (top_k=5). Deduplicate. Concat.

- **tool_node**

- Reads: sub_queries, route
- Writes: tool_result (str)

- Logic: Run arXiv API tool + DuckDuckGo search on each sub-query. Merge results.

- **answer_node**

- Reads: question, retrieved, tool_result, context_window, reflection_note
- Writes: answer (str), report (dict)

- Logic: Prompt LLM to synthesise from retrieved + tool_result only.

- **eval_node**

- Reads: answer, retrieved, tool_result, eval_retries
- Writes: faithfulness (float), eval_retries (int)

- Logic: Score overlap between answer tokens and retrieved context using BM25 or LLM-as-judge.

- **reflect_node**

- Reads: answer, question, faithfulness, retrieved
- Writes: reflection_note (str), answer (updated critique)

- Logic: LLM critique prompt — identify unsupported claims, suggest grounded revision.

- **save_node**

- Reads: answer, report, sources, messages
- Writes: messages (append), context_window (update)
- Logic: Persist final report as AIMessage. Update sliding window.

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07 RAG Knowledge Base Design

The ChromaDB knowledge base contains 45+ documents targeting the AI/ML Deep Tech research domain.

Source	# Docs	Content Type	Load Method
arXiv abstracts	20	Plain text abstracts	ArxivLoader
Attention Is All You Need	1	Full paper PDF	PyPDFLoader
LangChain / LangGraph docs	5	Markdown pages	WebBaseLoader
Hugging Face model cards	10	JSON → text	Custom loader
ARIA domain config	1	YAML structured docs	Custom parser
Research summaries	8	PDF abstracts	PyPDFLoader

Chunking Strategy:

```
from langchain.text_splitter import RecursiveCharacterTextSplitter
splitter = RecursiveCharacterTextSplitter(
    chunk_size=512,        # optimal for sentence-transformers
    chunk_overlap=64,      # preserve boundary context
    separators=['\n\n', '\n', '. ', ' ', '']
)
```

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08 Tool Registry

Tool Name	Type	Library	What It Does	Req?
`arxiv_search`	External API	langchain-community	Query arXiv by keyword; returns title, abstract	P0
`web_search`	Web Search	langchain-community	Live web search for news, blogs	P0
`citation_fmt`	Domain Tool	Custom	Format a list of sources into citations	P1
`domain_kb`	Vector Search	ChromaDB	Semantic search over ChromaDB	P0
`date_context`	Utility	Python	Return current date for grounding	P1
`pdf_reader`	File Tool	PyPDFLoader	Accept user PDF and add to Session	P2

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09 Self-Reflection & Faithfulness Gate

The self-reflection loop is ARIA's quality assurance mechanism, ensuring every answer is grounded in retrieved context before delivery.

```
# aria/nodes/eval_node.py
EVAL_PROMPT = '''
You are a faithfulness evaluator. Score how well the answer is grounded in the c
Rules:
- 1.0 = every claim is supported
- 0.7 = most claims are supported
- 0.5 = significant unsupported claims
- 0.0 = entire hallucination
'''
```

Evaluation Logic: If `score >= 0.7` → proceed to `save_node`. If `score < 0.7` → route to `reflect_node` for critique and re-generation via `answer_node` (max retries = 2).

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10 Conversation Memory Design

Using `MemorySaver` checkpointer, state persistence is keyed by Streamlit's unique `thread_id`. The sliding window guarantees the context window never exceeds LLM token limits, retaining the last 10 message pairs (`WINDOW_SIZE * 2` entries).

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11 Streamlit Deployment

The Streamlit UI enables non-technical users to run full research sessions. - **Sidebar**: Session management, domain selector. - **Thinking Expander**: Collapsible display of sub-queries. - **Report Card**: Structured JSON renderer for Summary, Findings, Sources. - **Faithfulness Badge**: Green/Amber/Red scores on quality. - **Source Explorer**: Citations dropdown.

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12 Tech Stack & Dependencies

Layer	Technology	Version	Purpose
Orchestration	LangGraph	>=0.2.0	StateGraph, MemorySaver
LLM	google-genai / groq	latest	Gemini Flash or Llama-3.3-70B
Embeddings	sentence-transformers	>=2.6	all-MiniLM-L6-v2
Vector DB	chromadb	>=0.5	Persistent local collection
UI	streamlit	>=1.35	Frontend on Streamlit Cloud
Type Safety	pydantic	>=2.0	Structured outputs

13 Mandatory Capability Compliance Matrix

#	Capability	ARIA Implementation	Status
1	LangGraph Workflow	7-node StateGraph: memory → planner → retrieve/tool → answer → eval → reflect	PASS
2	RAG Knowledge Base	45+ documents across 6 source types in ChromaDB; grounded synthesis	PASS
3	Conversation Memory	MemorySaver checkpointer per Streamlit session with sliding window	PASS
4	Self-Reflection	`eval_node` scores faithfulness 0-1, tracking LLM hallucinations and retrying	PASS
5	Tool Use	arXiv search, web search via DuckDuckGo, citation formatter	PASS
6	Deployment	Full Streamlit app with chat UI, report cards, and export via Streamlit Cloud	PASS

14 Implementation Roadmap

- **Week 1:** Foundation & State Design, ChromaDB initialisation.
- **Week 2:** Core Nodes: Planner + Retrieval, StateGraph wiring.
- **Week 3:** Tools + Self-Reflection Loop.
- **Week 4:** Memory + Streamlit UI, history persistence.
- **Week 5:** Polish, Testing & Deployment.

15 File Structure

```
aria/
├── app.py                        # Streamlit entry point
├── requirements.txt              # All dependencies pinned
├── config.yaml                   # Domain + model config
├── .env.example
├── aria/
│   ├── state.py                  # ARIAState TypedDict
│   ├── graph.py                  # build_graph()
│   ├── knowledge_base.py
│   ├── nodes/
│   ├── tools/
│   └── prompts/
├── data/
│   ├── raw/
│   └── chroma_db/
└── tests/
```

16 Risk Register

Risk	Likelihood	Impact	Mitigation
LLM API rate limits hit	Medium	High	Use generous free tiers + response cache
arXiv/DuckDuckGo empty	Low	Medium	Fallback to KB-only retrieval
Faithfulness loop stuck	Low	High	`eval_retries` cap at 2 → force pass flag
ChromaDB cold start delay	Medium	Medium	Pre-build and commit `chroma_db/` to repo